INF10025 Data Management and Analytics

Test 1 – Practice Questions

1. Getting started…

Below is a list of movies. You can create the same list if you create a query which only selects movies that contain the letter Z or the letter X. Save this query as NEWMOVIE. Now create queries (such as the ones below) based on the query named NEWMOVIE instead of the table named MOVIE.

MOVIENO	TITLE	RELYEAR	RUNTIME	RATINGCODE
421	The Life Aquatic With Steve Zissou	2004	119	MA
603	The Matrix	1999	136	MA
604	The Matrix Reloaded	2003	138	MA
605	The Matrix Revolutions	2003	129	MA
673	Harry Potter and the Prisoner of Azkaban	2004	141	PG
675	Harry Potter and the Order of the Phoenix	2007	138	M
745	The Sixth Sense	1999	107	M
1930	The Amazing Spider-Man	2012	136	M
4638	Hot Fuzz	2007	121	MA
9398	Zoolander	2001	89	M
11450	Quiz Show	1994	133	M
11584	Roxanne	1987	107	PG
19908	Zombieland	2009	88	MA
27578	The Expendables	2010	103	MA
36657	X-Men	2000	104	M
49051	The Hobbit: An Unexpected Journey	2012	169	M
50646	Crazy, Stupid, Love.	2011	118	M
68728	Oz: The Great and Powerful	2013	130	PG
72190	World War Z	2013	116	M
74465	We Bought a Zoo	2011	124	PG
76163	The Expendables 2	2012	103	MA
91314	Transformers: Age of Extinction	2014	165	PG
102382	The Amazing Spider-Man 2	2014	142	M
109445	Frozen	2013	102	G
127585	X-Men: Days of Future Past	2014	131	M
329833	Zoolander 2	2016	100	M

2. Following on from the above, how many rows are selected by each of these queries?

Field:	MOVIENO	TITLE	RELYEAR	RUNTIME
Table:	newmovie	newmovie	newmovie	newmovie
Sort:				
Show:	☑	☑	☑	☑
riteria:		Like "*j*"		
or:		Like "*k*"		

Field:	MOVIENO	TITLE	RELYEAR	RUNTIME
Table:	newmovie	newmovie	newmovie	newmovie
Sort:				
Show:	☑	☑	☑	☑
Criteria:		Like "*g*" Or Like "*f*"		
or:				

Field:	MOVIENO	TITLE	RATINGCODE	RELYEAR
Table:	newmovie	newmovie	newmovie	newmovie
Sort:				
Show:	☑	☑	☑	☑
Criteria:			"PG"	
or:			"MA"	

Field:	MOVIENO	TITLE	RATINGCODE
Table:	newmovie	newmovie	newmovie
Sort:			
Show:	☑	☑	☑
Criteria:			"G" Or "M"
or:			

Field:	MOVIENO	TITLE	RATINGCODE
Table:	newmovie	newmovie	newmovie
Sort:			
Show:	☑	☑	☑
Criteria:			"G" And "PG" And "MA"
or:			

Field:	MOVIENO	TITLE	RUNTIME
Table:	newmovie	newmovie	newmovie
Sort:			
Show:	☑	☑	☑
Criteria:			>=100 And <=110
or:			

Field:	MOVIENO	TITLE	RATINGCODE	RELYEAR
Table:	newmovie	newmovie	newmovie	newmovie
Sort:				
Show:	☑	☑	☑	☑
Criteria:			"M"	>2010
or:				

Field:	MOVIENO	TITLE	RATINGCODE	RELYEAR
Table:	newmovie	newmovie	newmovie	newmovie
Sort:				
Show:	☑	☑	☑	☑
Criteria:			"M"	>2010
or:			"PG"	<2000

Field:	MOVIENO	TITLE	RATINGCODE	RELYEAR
Table:	newmovie	newmovie	newmovie	newmovie
Sort:				
Show:	☑	☑	☑	☑
Criteria:			"MA"	>2010
or:				<2000

Field:	MOVIENO	TITLE	RATINGCODE	RELYEAR	RUNTIME
Table:	newmovie	newmovie	newmovie	newmovie	newmovie
Sort:					
Show:	☑	☑	☑	☑	☑
Criteria:			"PG"	>=2010	>100
or:					

Field:	MOVIENO	TITLE	RATINGCODE	RELYEAR	RUNTIME
Table:	newmovie	newmovie	newmovie	newmovie	newmovie
Sort:					
Show:	☑	☑	☑	☑	☑
Criteria:			"G"		
or:				>=2013	
					<95

Field:	MOVIENO	TITLE	RUNTIME	RELYEAR
Table:	newmovie	newmovie	newmovie	newmovie
Sort:				
Show:	☑	☑	☑	☑
Criteria:			>=100 And <=110	<2000 Or >2010
or:				

If you want more questions like these – make your own. Simply follow the styles above.

3. How many rows are generated by each query that follows (note: count)?

Field:	RATINGCODE	RATINGCODE
Table:	MOVIE	MOVIE
Total:	Group By	Count
Sort:		
Show:	☑	☑

Field:	RELYEAR	RELYEAR
Table:	MOVIE	MOVIE
Total:	Group By	Count
Sort:		
Show:	☑	☑
Criteria:		
or:		

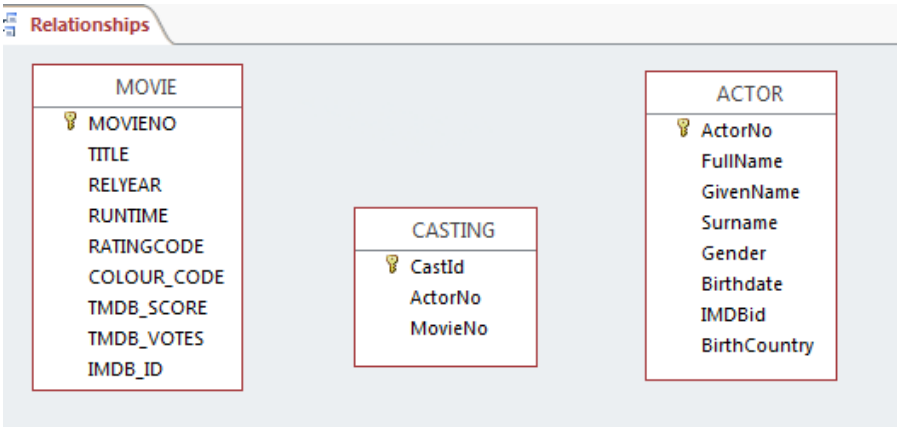
Field:	RELYEAR	RATINGCODE	RATINGCODE	RELYEAR
Table:	MOVIE	MOVIE	MOVIE	MOVIE
Total:	Group By	Group By	Count	Where
Sort:				
Show:	☑	☑	☑	☐
Criteria:				>=2014
or:				

4. Answer the following true/false and short answer questions..

- A PK may have a null value? T/F
- A FK may have a null value? T/F
- Every table requires a PK? T/F
- Every table requires a FK? T/F
- How many PKs can a single table have?
- How many FKs can a single table have?
- A table can have both a PK and a FK? T/F
- A table may have neither a PK and a FK? T/F
- Does every FK value in a table need to be unique?
- Does every PK value in a table need to be unique?
- Does every FK value in a database of many tables need to be unique?
- Does every PK value in a database of many tables need to be unique?
- What is the purpose of an AutoNumber datatype?
- Can a table have more PKs than FKs?
- Can a table have more FKs than PKs?
- Can a table have zero PKs?
- Can a table have zero FKs?
- Can a table have zero Autonumber fields?
- What is a query parameter?
- Can a query have more than one parameter?
- When creating fields in a table, you must specify a datatype. What are the commonly used datatypes?
- What is a CSV file?
- What is a delimiter?
- What is a fixed format file?
- What is an index?
- What is the benefit of using an index?
- What is the cost of using many indexes within a single table?
- You are planning to create a table to store the fields: ‘Pet Name’; ‘Pet Purchase Price’; ‘Pet Birthday’; ‘Pet Number of feet’; and ‘Pet Has a Microchip’. Which data type would you choose for each field? Justify your reasons.
- How does a query differ from a table?
- Can you use a table as the source data for a query?
- Can you use two tables as the source data of a query?
- Can you use another query as the source data for a query?

5. Finally, a few questions about 'relationships'…

- In the relationship manager, what does Enforce Referential Integrity mean?
- How many PKs are in the diagram below?
- Draw the M:1 relationships between these tables.
- How many relationship lines have you drawn?
- How many foreign Keys exist in the Movie table?
- How many foreign Keys exist in the Actor table?
- How many foreign Keys exist in the Casting table?



Note: no answers are provided for these practice questions – you will need to run the queries in MS Access or Libre Base.